



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251 L1

Spring Semester 2025

Instructor: Usman Salahuddin

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Office: C-103

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Research Project (30%)

- Group activity (Groups of 3)
- Students review literature and conduct field work to understand local water related problems and propose possible solutions.

	Grade % (Total 30%)
Proposal	2.5
Introduction	2.5
Literature review	2.5
Primary research	7.5
Presentation	5
Final Report	10

Proposal (2.5%)

- Term paper proposal
- Format (Times New Roman, Font Size 11, Heading size 14, Normal margins, single line spacing, IEEE style)
- Topic (different topic for each group)

Names

Enrollment

Title

Background

Objectives/Goal

Methods

Expected contribution to the society

References

Title

Bad example: “Water for Karachi: Innovating Solutions to Alleviate Water Scarcity and Improve Urban Resilience”

Good example: "Karachi's Water Crisis: Finding Ways to Fix Water Shortage"

Intro

Short background on what the problem is, why it is important, why it's not been solved till yet, etc.

Objectives

Bullet points, should be practical, straightforward and concise

Methodology

Thought process of how you are going to do it, what tools will you use, what resources, etc.

Expected contribution to the society

What problem will it solve, how it is going to impact the society

References

IEEE style, font size 9

Water Journal (15%)

- Students keep a log of their daily water usage and then use that data to calculate water footprints

Activity	Average Water Usage (liters)
Drinking water	
Showering	
Washing hands	
Cooking	
Flushing toilet	
Brushing teeth	
Laundry	
Other (specify)	
Virtual water (products)	
Total Daily Usage	

Deadlines

- Water Journal (Jan 21)
- Project proposal (Jan 28)

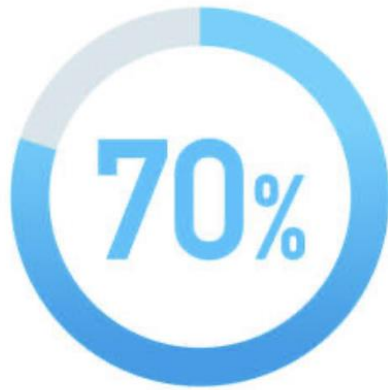
Where did water come from?

- It is He who sends down water for you from the sky, from which comes a drink for you, and the shrubs that you feed to your animals... —*from The Prophets (21:30)*.
- The earth was formless and void, and darkness was over the surface of the deep, and the Spirit of God was moving over the surface of the waters. — **Gen. 1:2**.
- Scientific view

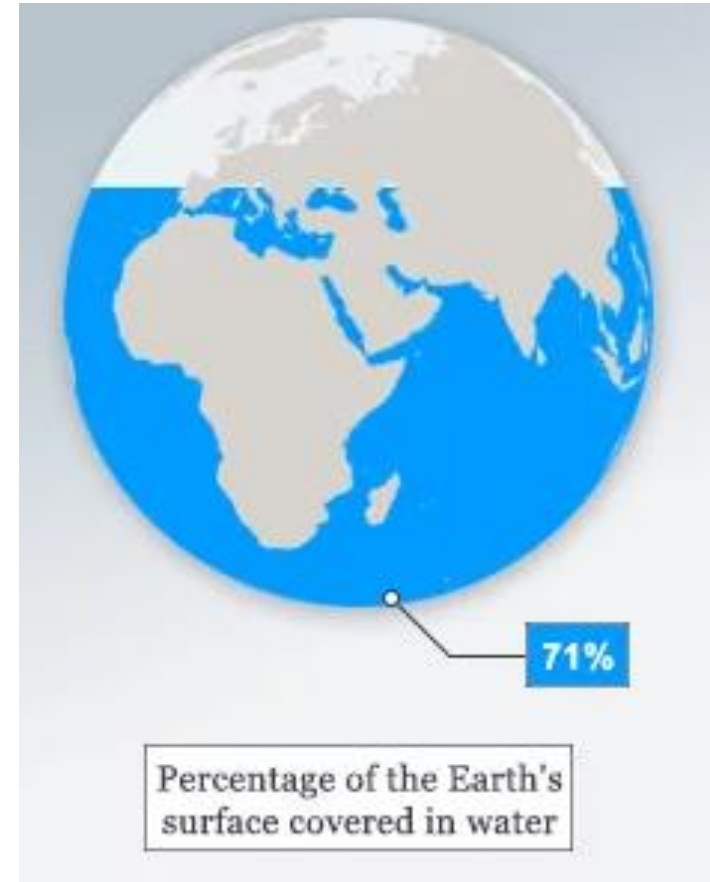
WATER

*"Water is H₂O, hydrogen two parts, oxygen one, But there is also a third thing, that makes it water, And nobody knows what it is.
—D. H. Lawrence, "The Third Thing"*

HUMAN BODY WATER LEVEL



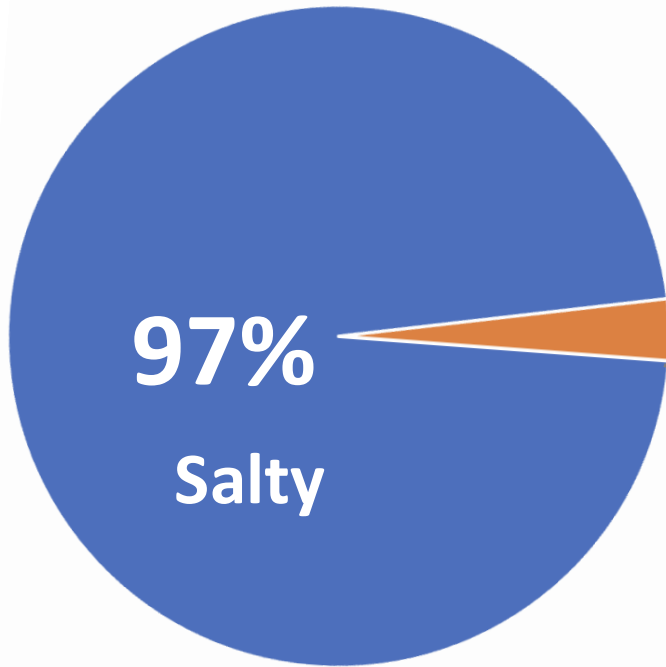
Source: istockphoto.com



Source: USGC, DW

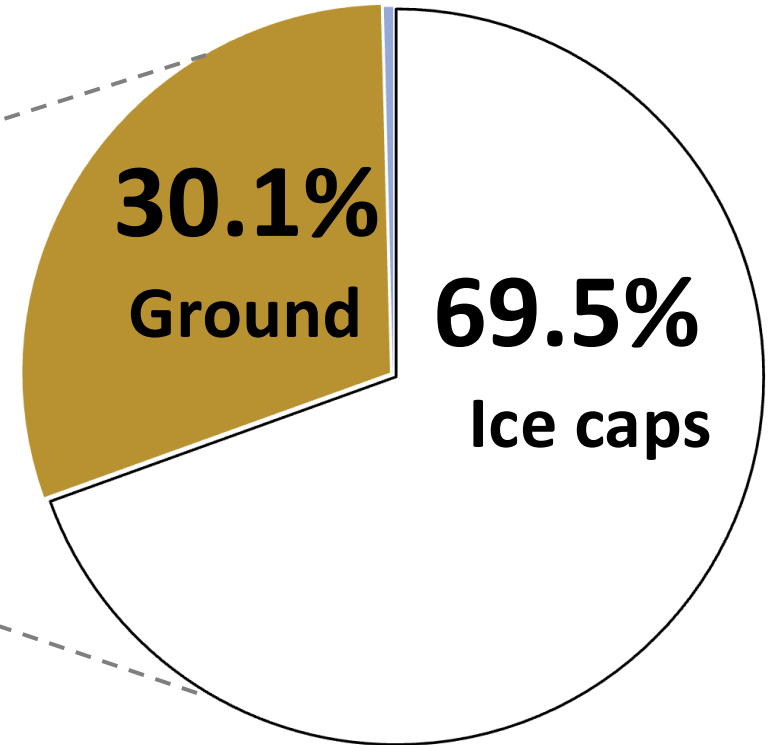
WATER

Total Earth's water



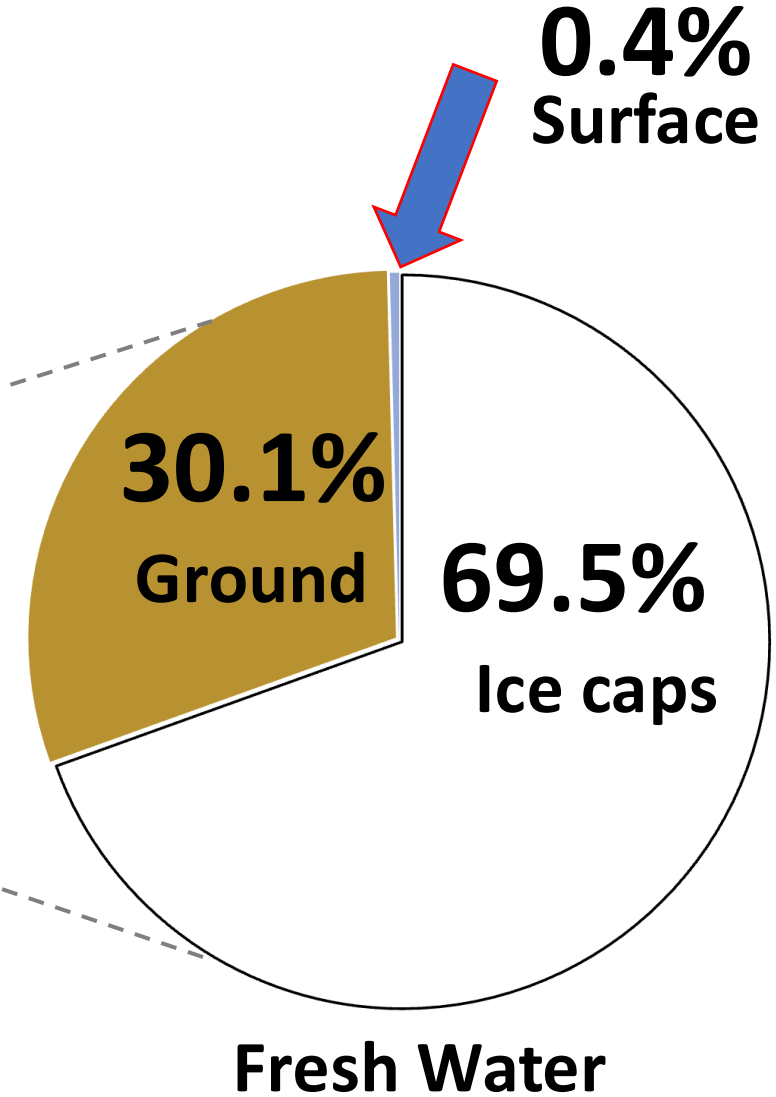
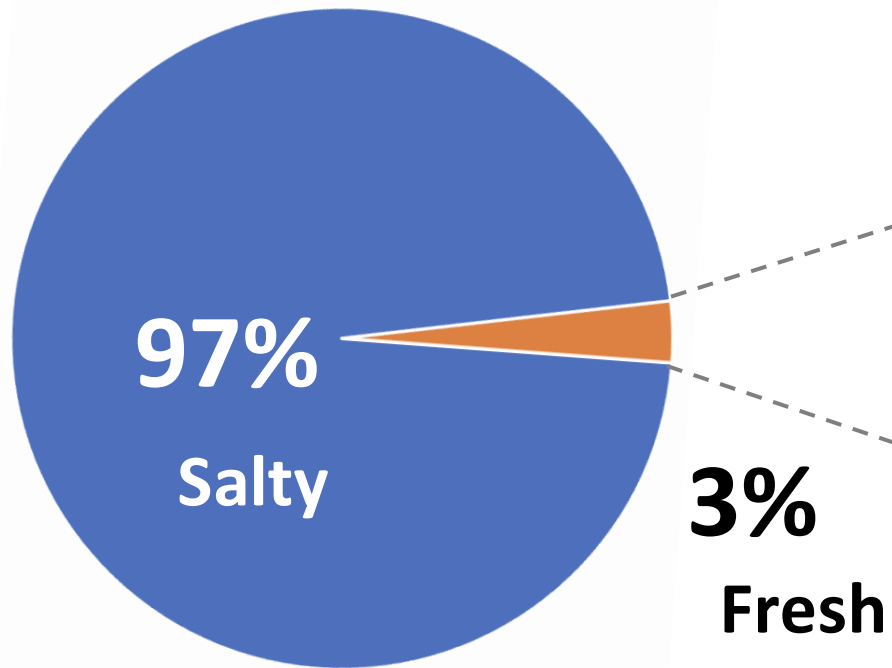
3%
Fresh

Fresh Water

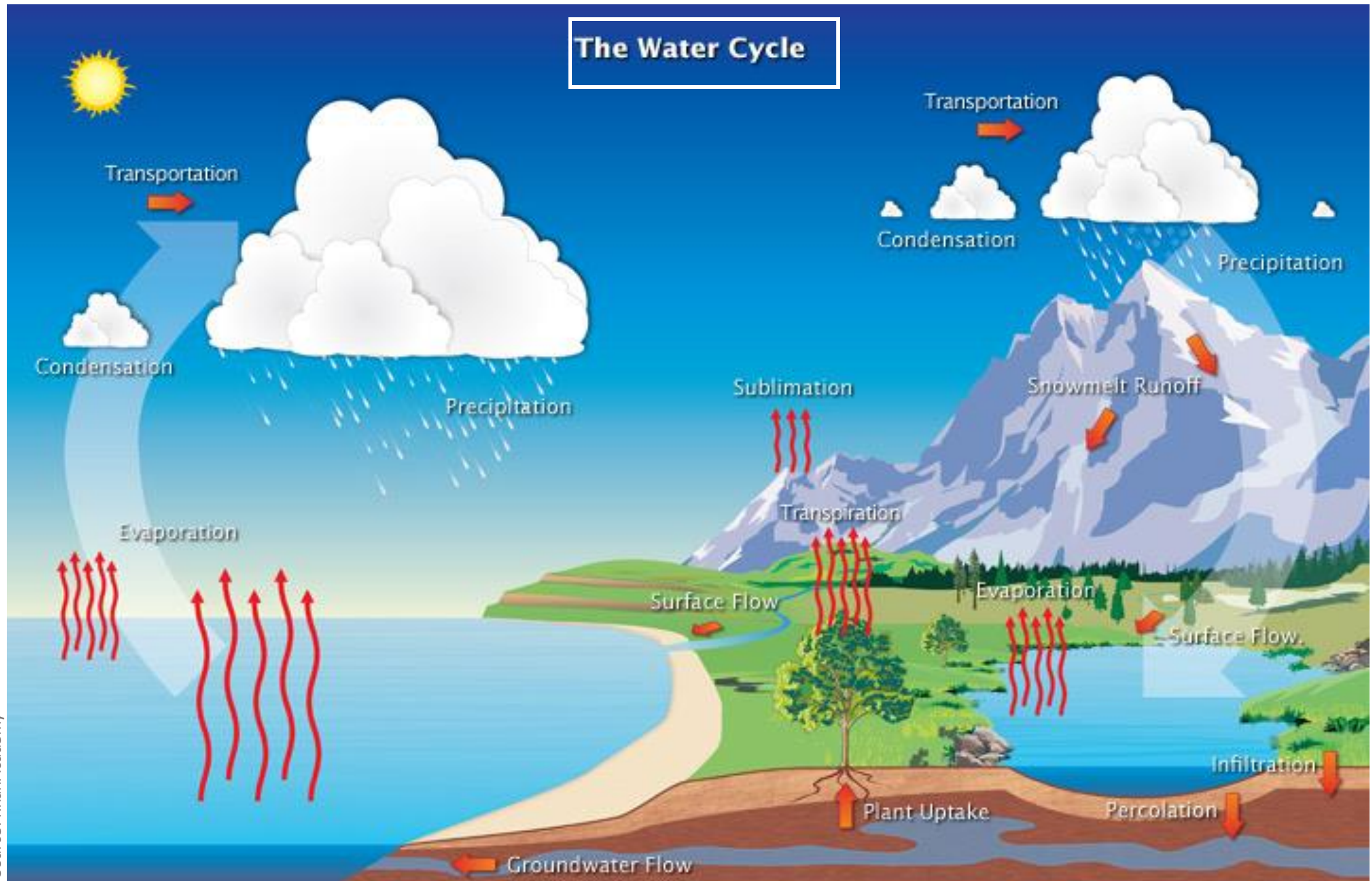


WATER

Total Earth's water



The Water Cycle



WATER ACCESS

World Population & Income levels as of 2017

			
LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4
\$2	\$8	\$32	
• Drinking water: Mud-holes • Transport: Walk barefoot • Cooking: Using firewood • Food: Eat what you grow/find • Other: High child mortality rates (can't afford antibiotics)	• Drinking water: Buckets • Transport: Sandals, bicycles • Cooking: Small gas stove • Food: Buy some food/rear chicken • Other: Unstable electricity, light bulbs	• Drinking water: Outdoor tap • Transport: Motorcycles • Cooking: Proper stove • Food: Fridge, variety of dishes • Other: Can afford medical bills & high school education for kids	• Drinking water: Hot/cold water indoors • Transport: Cars, airplanes • Cooking: Full kitchen • Food: Huge variety • Other: >12 years of education

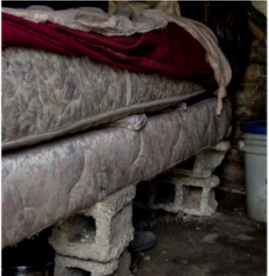
 = 1 bil people

Income: per person per day, adjusted for price differences

Source: Gapminder

WATER ACCESS

LIFE ON THE FOUR INCOME LEVELS

	LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4		LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4	
COOKING					DRINKING WATER					
EATING					TRANSPORTATION					
SLEEPING										



Spring of 2008 (Barcelona)

- Drought in Catalonia for 18 months
- Barcelona (5M population)
- Reservoirs 80% empty
- 37 miles line from nearby river, Cost (\$280M)
- Desalination plant

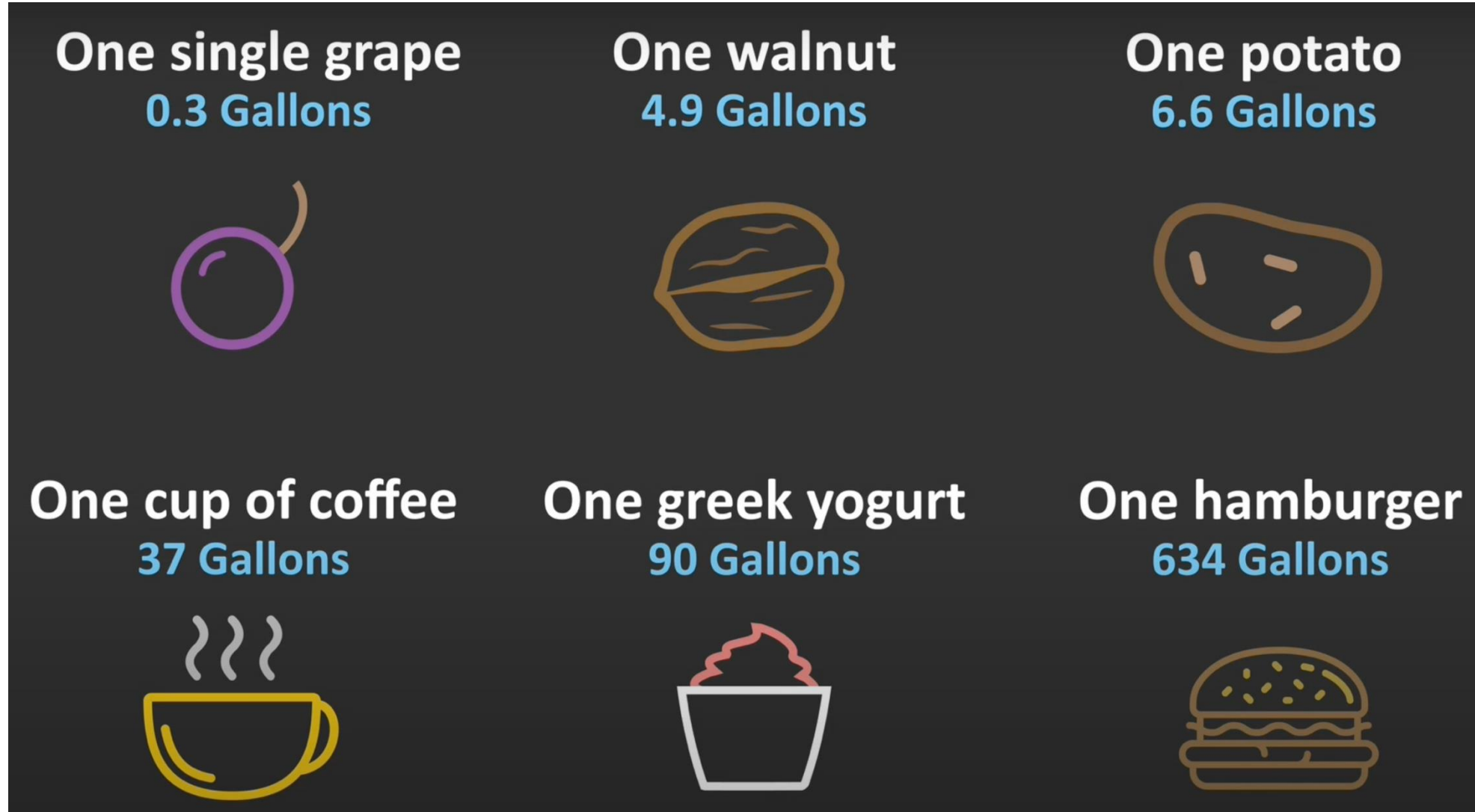
Spring of 2008

- \$30M for water supply via ships over summer
- 5M gallons of water delivered by the 1st ship.



Source: CSDefender.com

Water is invisible



Source: S. Darling et al.

Water is invisible



One pound of steel
11 Gallons



One pound of cement
0.25 Gallons



One pound of lumber
1.7 Gallons



One pound of plastic
24 Gallons



One pound of paper
39 Gallons

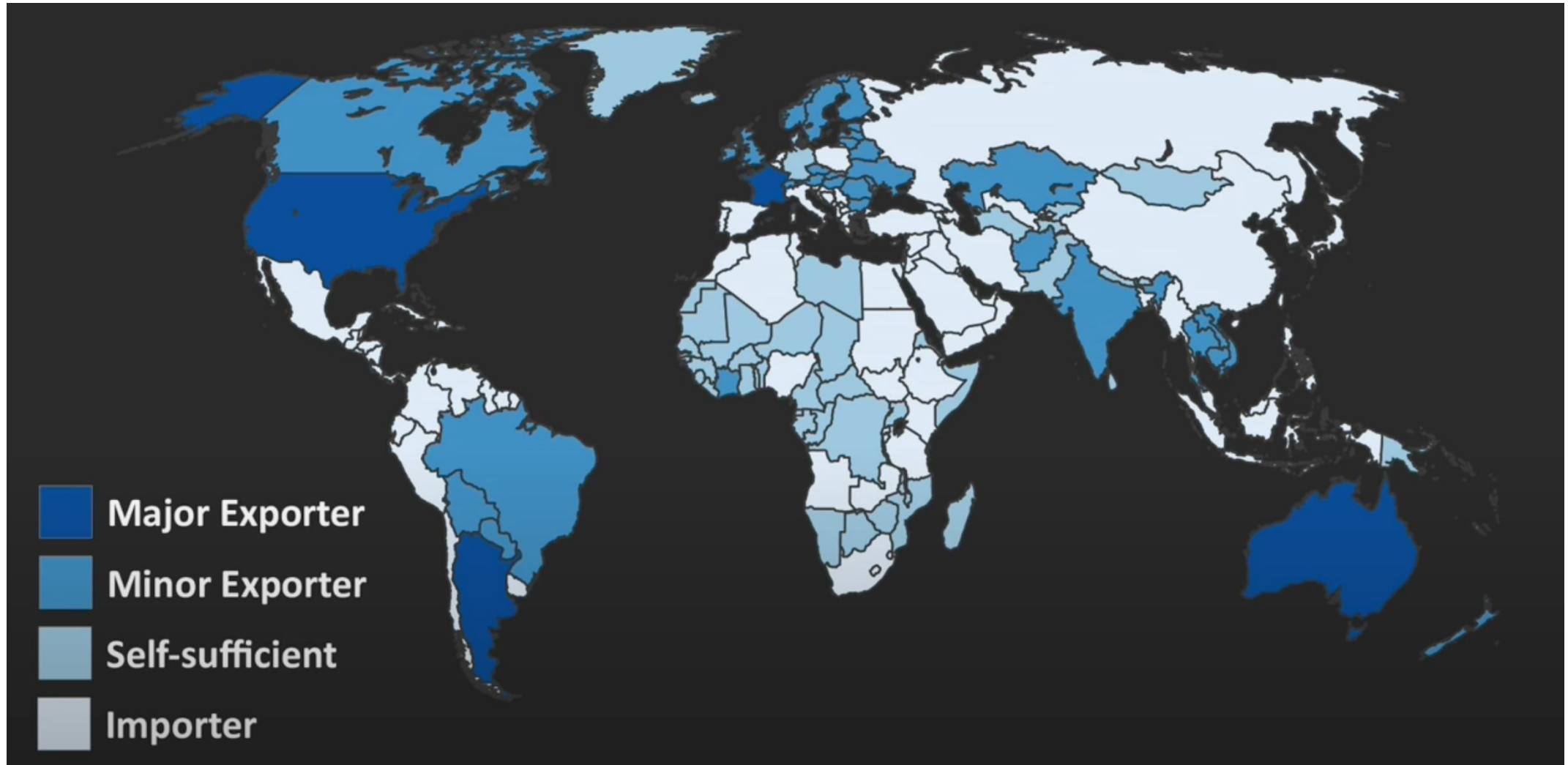


One pound of rubber
60 Gallons



One pound of cotton
101 Gallons

WATER TRADE MAP



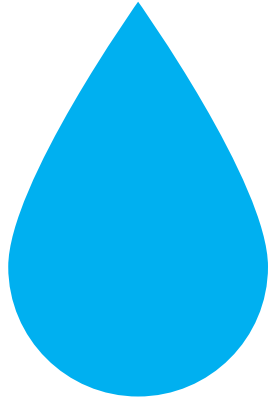
Source: S. Darling et al.

Water footprint Video

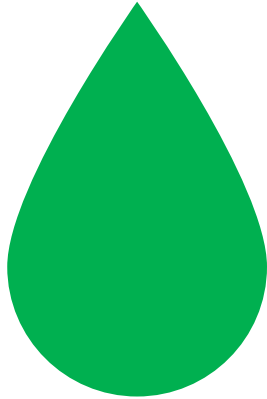
WATER HAS FLAVORS!



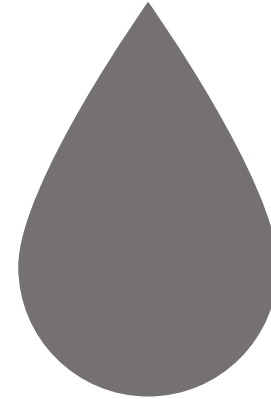
WATER HAS FLAVORS!



**Lakes,
Streams
Underground**



**Natural rain
Plants
Soil**



**Used
Polluted
Wastewater**

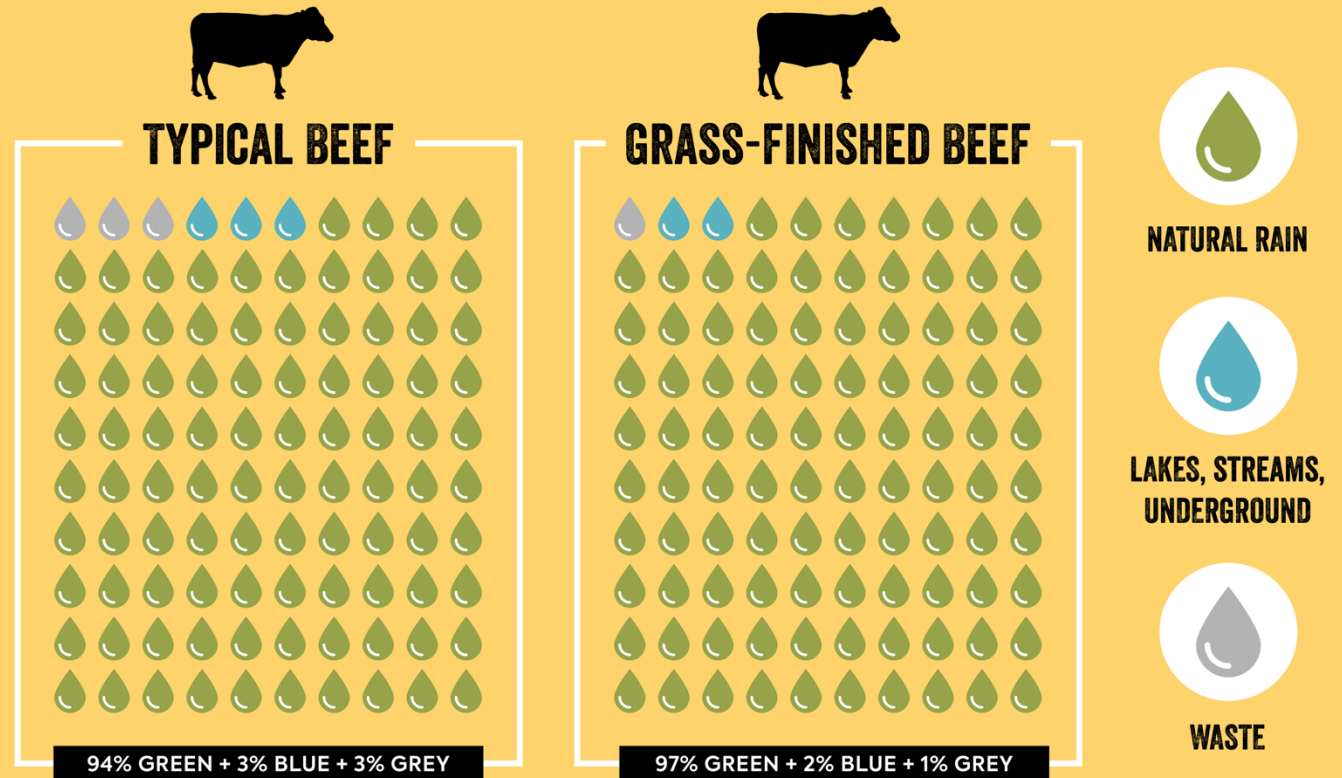
WATER FOOTPRINT OF BEEF

- 1847 gallons to produce 1 lb of beef
- Cows still emit CO₂ though
- Monocrop agriculture
- Beef is necessary

BEEF ISN'T A WATER HOG

@SUSTAINABLEDISH | SACREDCOW.INFO

Most of the droplets below represent "green water", or **natural rainfall**.¹ "Blue water" represents water that has been sourced from surface or groundwater resources. **Beef requires only 280 gallons²** of "blue water" per pound, which is **less than** the amount required to produce a pound of avocados, walnuts, or sugar. Grey water is the volume of water required to dilute pollutants.



1. <https://waterfootprint.org/media/downloads/Report-48-WaterFootprint-AnimalProducts-Vol1.pdf>
2. <https://www.sciencedirect.com/science/article/pii/S0308521X18305675>

MEATLESS MONDAYS

GLOBAL LIVESTOCK PRODUCTION
CREATES MORE GREENHOUSE
GAS THAN THE ENTIRE
TRANSPORTATION SECTOR.



#MeatlessMonday  MeatlessMonday.com

DECREASE YOUR CHANCE OF
GETTING DIABETES BY ABOUT **15%**



Just skip a serving of meat every day and replace it
with a vegetable protein like black beans or tofu.

#MeatlessMonday  MeatlessMonday.com

YOUR KIDNEYS
KEEP YOU ALIVE.



THE LEAST
YOU CAN DO
IS SERVE THEM
A GOOD MEAL.

Choosing Meatless Monday and eating plant-based
proteins instead of meat can help maintain kidney health.
So feed them like you need them. Because you do.

  National Kidney Foundation* [MeatlessMonday.com](https://meatlessmonday.com)

DYNAMICS of WATER

- All water problems are somewhat local
- We are using less water now than before
- Water footprint idea is misleading (its not like carbon footprint)
- Water is indestructible and recyclable